Pre-Lab 01:
Frequency Response of Operational Amplifiers

ECE 1212: Electronic Circuit Design Lab

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Team Roles:

Activity	**Student Name**
Circuit Construction	Stefan Zhang
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Report Writing	Stefan Zhang; Nick Zhu

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1) Pre-Lab (1A)

1.1) Question 1: Non-Inverting Amplifiers

Using the typical LM741 gain-bandwidth product,

$$A_V f_{3\text{dB}} \approx f_t, \qquad f_t \approx 1.0\,\text{MHz}.$$

1.1.1) $A_V = 1$ Voltage Follower

The minimum-component design is a voltage follower.

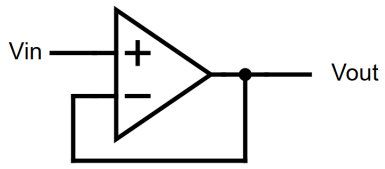


Figure 1: Non-inverting voltage follower with $A_V = 1$

For $A_V = 1$,

$$f_{3\text{dB}} \approx \frac{f_t}{A_V} = 1.0\,\text{MHz}.$$

The voltage follower acts as a high-input-impedance, low-output-impedance buffer that reduces loading.

1.1.2) $A_V = 10$ Non-Inverting Amplifier

For a non-inverting amplifier,

$$A_V = 1 + \frac{R_2}{R_1} = 10.$$

Choosing

$$R_1 = 3\,\text{k}\Omega, \qquad R_2 = 27\,\text{k}\Omega$$

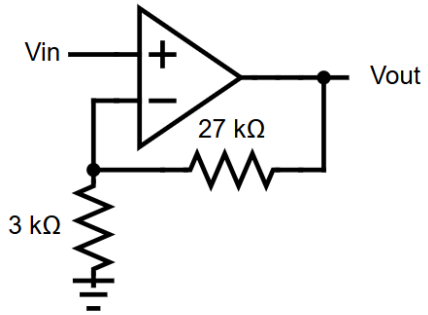


Figure 2: Non-inverting amplifier with $A_V = 10$

$$f_{3\mathrm{dB}} \approx \frac{1.0\,\mathrm{MHz}}{10} = 100\,\mathrm{kHz}.$$

1.2) Question 2: Input Impedance of the Non-Inverting Amplifier

At $f = 1\,\mathrm{kHz}$, insert a known series resistor R_s, measure V_s and V_{in}, and calculate

$$Z_{in} = R_s \frac{V_{in}}{V_s - V_{in}}.$$

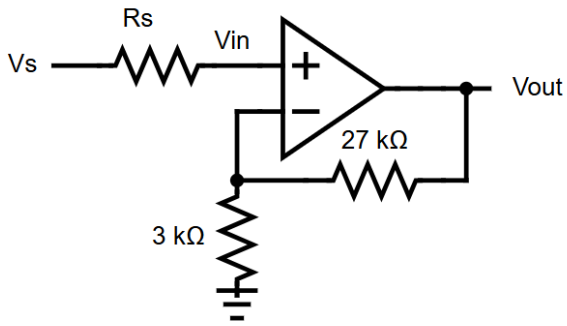


Figure 3: Input-impedance measurement circuit for the non-inverting amplifier

Using $r_i \approx 2\,\mathrm{M\Omega}$,

$$R_{in,f} \approx r_i(1 + A\beta), \qquad |A| \approx \frac{f_t}{f} = 1000 \quad (f = 1\,\mathrm{kHz}).$$

Thus, the predicted input impedances are approximately

$$A_V = 1: \quad R_{in,f} \sim 2\,\mathrm{G\Omega},$$
$$A_V = 10: \quad R_{in,f} \sim 200\,\mathrm{M\Omega}.$$

These impedances are difficult to measure because they are much larger than the oscilloscope input impedance, so the measurement equipment significantly loads the circuit.

2) Pre-Lab (1B)

2.1) Question 1: High-Gain Inverting Amplifier

The design requirements are $R_{in} = 1\,\mathrm{k\Omega}, \quad R_L = 3.3\,\mathrm{k\Omega}, \quad f_{3\mathrm{dB}} \geq 20\,\mathrm{kHz}.$

For an inverting amplifier,

$$A_V = -\frac{R_f}{R_{in}}, \qquad A_{\mathrm{NG}} = 1 + \frac{R_f}{R_{in}}, \qquad f_{3\mathrm{dB}} \approx \frac{f_t}{A_{\mathrm{NG}}}.$$

Thus,

$$R_f \le R_{in}\left(\frac{f_t}{20\,\text{kHz}} - 1\right).$$

Using the nominal LM741 value $f_t = 1.00\,\text{MHz}$, $R_f \le 49\,\text{k}\Omega$. Therefore,

$$A_V = -47, \qquad f_{3\text{dB}} \approx \frac{1.00\,\text{MHz}}{48} = 20.8\,\text{kHz}.$$

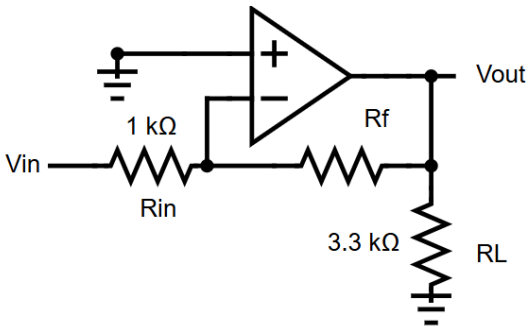


Figure 4: High-gain inverting amplifier

The closed-loop frequency response is

$$\frac{V_{out}}{V_{in}} = -\frac{R_f/R_{in}}{1 + \dfrac{1 + R_f/R_{in}}{A(jf)}}.$$

2.2) Question 2: DC Compensation

A DC input offset of up to $\pm 1\,\text{V}$ can drive the high-gain amplifier into saturation. An adjustable compensation voltage V_C is therefore applied to the summing node.

Choose

$$R_{comp} = R_{in} = 1\,\text{k}\Omega.$$

The V_{comp} is obtained from a $10\,\text{k}\Omega$ potentiometer connected between the $\pm 12\,\text{V}$ supplies.

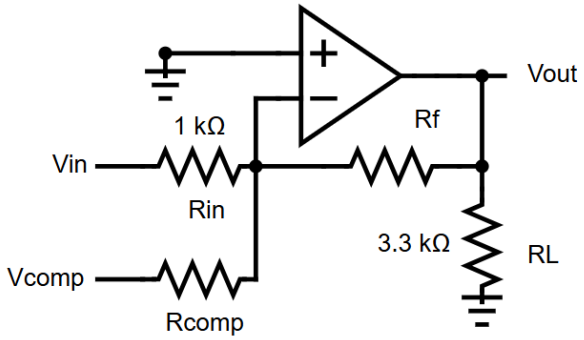


Figure 5: Inverting amplifier with adjustable DC compensation

For the DC component,

$$V_{out,DC} = -R_f \left(\frac{V_{in,DC}}{R_{in}} + \frac{V_{comp}}{R_{comp}} \right).$$

Setting $V_{out,DC} = 0$ gives

$$V_{comp} = -\frac{R_{comp}}{R_{in}} V_{in,DC} = -V_{in,DC}.$$

The potentiometer adjusts V_{comp} to cancel the DC offset and prevent output saturation.

2.3) Question 3: Input Impedance of the Inverting Amplifier

Using the same voltage-divider method,

$$Z_{in} = R_s \frac{V_{in}}{V_s - V_{in}}.$$

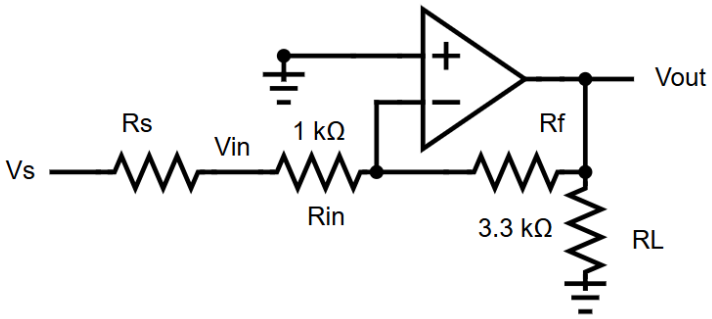


Figure 6: Input-impedance measurement circuit for the inverting amplifier

The same voltage-divider procedure will be repeated over frequency.

At low frequency, $Z_{in} \approx R_{in} = 1\,\text{k}\Omega$. With finite open-loop gain,

$$Z_{in}(jf) = R_{in} + \frac{R_f}{1 + A(jf)}, \qquad A(jf) \approx \frac{f_t}{jf}.$$

As frequency increases, $|A|$ decreases, so the virtual-ground condition weakens and $|Z_{in}|$ deviates increasingly from $1\,\text{k}\Omega$.